Counting Sort:

I- $\theta(n+k)$ time.

If $k \in \theta(n)$ then $I = \theta(n)$ time.

(linear)

Counting Sort is linear provided the largest entry given by user is bounded above by cn for some constant $c < n$.

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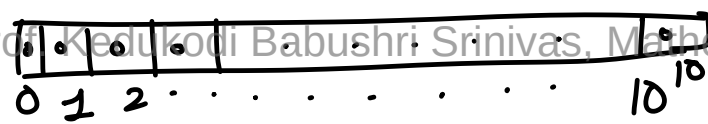
Worst input for Counting sort.

A	5	2	10^{10}

Note: $n = 3$. , $n^2 = 9$

k = largest number in A $= 10^{10} >> n^2 = 9$

In this case, the auxillary array C



(Huge space)

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Radix Sort:

e.g. Input:

1	4	2
2	5	1_a
2	4	1_b

stability

$d = 3$ digits.

$d = 3$ passes

1st Pass

2	5	1_a
2	4_a	1_b
1	4_b	2

2nd Pass

2	4_a	1
1	4_b	2
2	5	1

3rd Pass

1	4	2
2_a	4	1
2_b	5	1

Output

Sort:

$$
\begin{array}{l}
00|100|10 \\
0|10|10|001 \\
0|0|0001
\end{array}
\xrightarrow{\text{octal}}
\begin{array}{l}
142 \\
251 \\
241
\end{array}
$$

↓ Radix Sort

$$
\begin{array}{l}
142 \\
241 \\
251
\end{array}
$$

↓ Binary

$$
\begin{array}{l}
00|100|10 \\
0|0|0001 \\
0|10|10|001
\end{array}
$$

Octal Numbers

000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

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Radix Sort (A, d): (Array, digits)

1. For $i \leftarrow 1$ to d
2. do use a stable sort to sort array A on digit i.

Note: the above algorithm is written in an abstract way. Stable sort ✓ (counting sort eg).

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Result 1:

Given n-digit numbers in which each digit can take k possible values, Radix Sort sorts these numbers in $\theta(d(n+k))$ time.

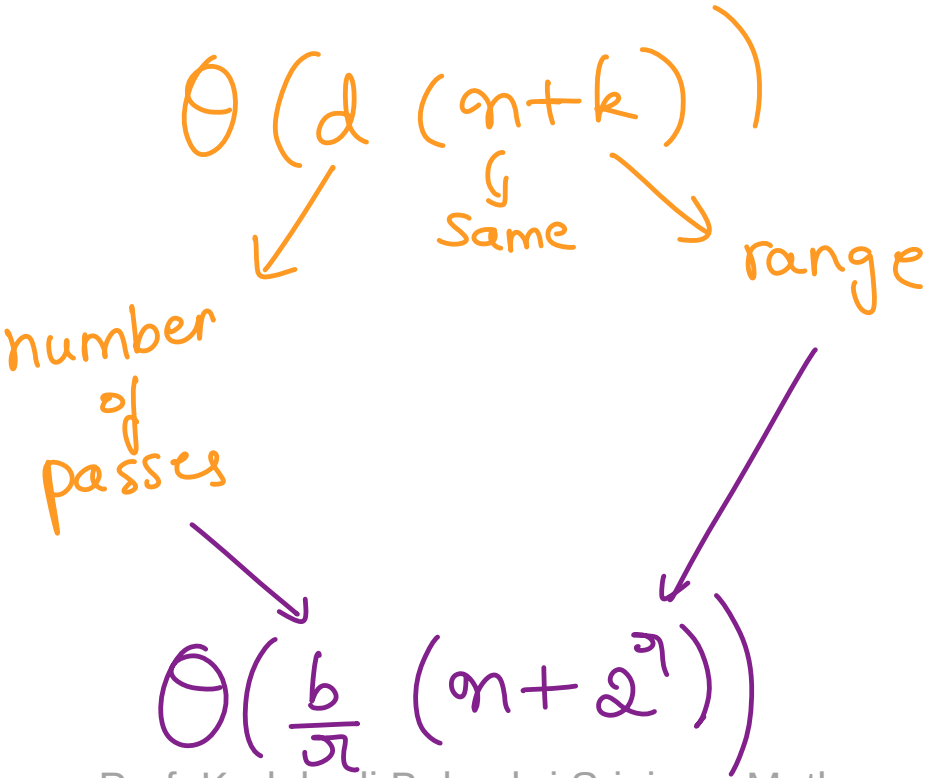
Proof.

A good stable sort (for example, counting sort) takes $\theta(n+k)$ time to process every column.

There are d passes (or columns).

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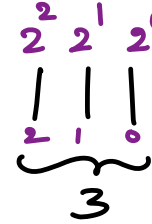
$\Rightarrow$ Total time is $\Theta(d(n+k))$ time.



Range:

n bits $\rightarrow$ Largest number

$n=3$



$(7 = 2^2 \times 1 + 2^1 \times 1 + 2^0 \times 1)$

n bits



$$2^{n-1} \times 1 + 2^{n-2} \times 1 + \cdots + 2^0 \times 1$$

(Geometric series with $c \cdot r = 2$)

$$= \frac{2^n - 1}{2 - 1} = 2^{n-1} \in \Theta(2^n)$$

$\frac{b}{r} \Leftrightarrow$ No of passes.

eg: $\overbrace{001}^{r=3}\overbrace{100}^{r=3}\overbrace{010}^{r=3}$

9 bits

$r = 3$ chosen. (9 is largest)

$$\frac{b}{r} = \frac{9}{3} = 3 = \text{No of passes} = d.$$

(Floor or ceiling is chosen accordingly if required).

Result 2: Given n b-bit numbers and any +ve integer $r \leq b$, Radix Sort sorts these numbers in

$$\Theta\left(\frac{b}{r}(n+2^r)\right) \text{ time.}$$

Proof:

Radix Sort takes $\Theta(d(n+k))$ time

(By Result 1)

k = largest number in n bits

$$= \underset{n-1}{1}\ \underset{n-2}{1}\ 1 \cdots 1\ \underset{1}{1}\ \underset{0}{1}$$

$$= 2^{n-1} \times 1 + 2^{n-2} \times 1 + \cdots + 2^{1} \times 1 + 2^{0} \times 1.$$

$$= \frac{2^n - 1}{2 - 1}$$

$$= 2^n - 1 \in \Theta(2^n)$$

d = No of passes

= no of digits

$$= \left\lceil \frac{\text{Total bits}}{r} \right\rceil = \left\lceil \frac{b}{r} \right\rceil$$

Total time $= \Theta(d(n+k))$

$= \Theta\left(\left\lceil \frac{b}{r} \right\rceil (n+2^r-1)\right)$

$= \Theta\left(\left\lceil \frac{b}{r} \right\rceil (n+2^r)\right)$ time.

n must be chosen

Minimize $f(n) = \frac{b}{n}(n+2^n)$

$$f(x) = \frac{b}{x}(n+2^x)$$

Let $f'(x) = 0$, $x = ?$

Aim is to get $\theta(f(n)) = \theta(n)$ by choosing good r.

Want: $\frac{b}{r}(n + 2^r) \in \theta(n)$

$\Rightarrow n + 2^r \in \theta(n)$

$\Rightarrow 2^r \leq n$.

$\Rightarrow r = \lceil \log_2 n \rceil$

$$T = \Theta\left(\frac{b}{n}\left(n + 2^n\right)\right)$$

$$= \Theta\left(\frac{b}{\log_2 n}\left(n + 2^{\log_2 n}\right)\right)$$

$$= \Theta\left(\frac{b}{\log_2 n}\left(n + n\right)\right)$$

$$= \Theta\left(\frac{b}{\log_2 n} \cdot n\right)$$

$$= \Theta(n) \text{ if } \frac{b}{\log_2 n} \in \Theta(1)$$

(Almost linear)

Req: cn form. c - constant & $c \in \Theta(1)$

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Case 1 : $a < b < \lfloor \log_2 n \rfloor$

Case 2 : $b \geq a \geq \lfloor \log_2 n \rfloor$

$n = 10$, $b = 9$.

$a = \lfloor \log_2 10 \rfloor \geq \log_2 8 = 3$. (Good choice)

$$\frac{b}{\log_2 10} \leq \frac{9}{3} = 3 \in \Theta(10)$$

$n=3. \quad b=9$

$a = \lfloor \log_2 3 \rfloor \geq \log_2 2 = 1.$

$$\frac{b}{\log_2 3} \leq \frac{9}{2}.$$

Case 1: $a < b < \lfloor \log_2 n \rfloor$

$$\theta\left(\frac{b}{a}(n + 2^a)\right)$$

$$= \theta\left(\frac{b}{a}(n+n)\right)$$

$2^a \leq 2^{\log n} \leq n.$

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$= \Theta\left(\frac{b}{a} n\right)$

$= \Theta(n)$ if $\Theta\left(\frac{b}{a}\right) \in \Theta(1)$

$a = n$, $\frac{b}{n} \stackrel{?}{\in} \Theta(1)$.